Graphs & Transformations

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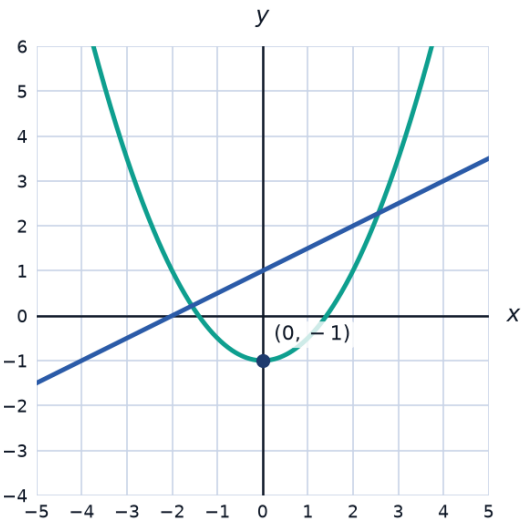
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Basic Graph Forms & Translations

The graph of $y = f(x)$ is a curve passing through the points A(1, 2), B(4, 2), and C(2, 5). This graph is then subjected to a sequence of transformations. First, it is reflected in the line $y = x$. Let the image of the points A, B, and C under this reflection be A', B', and C' respectively. Secondly, the graph of the reflected curve is translated by the vector $(-3; 0)$. Let the final image of the points A', B', and C' after this translation be A'', B'', and C''. Given that the original curve is a parabola with its axis of symmetry parallel to the y-axis, and passes through A, B, and C, find the coordinates of C''.

2 [Foundational] • Target: Reflections and Stretches

The diagram shows the graph of the function $y = f(x)$. The graph is transformed by a sequence of operations. The resulting graph, $y = g(x)$, is also shown. The function $f(x)$ is given by $f(x) = ax^2 + bx + c$, where $a, b,$ and c are constants. The vertex of the graph of $y = f(x)$ is at $(2, -1)$, and it passes through the point $(4, 3)$. The graph of $y = g(x)$ is obtained by translating the graph of $y = f(x)$ by $(-1\ 3)$ followed by a reflection in the y-axis. Find the values of $a, b,$ and c.



3 [Foundational] • Target: Composite Transformations

The diagram shows the graph of the function $y = f(x)$. The coordinates of the points A, B, and C are given. The graph is transformed by a translation of 3 units to the right and 2 units down. What are the new coordinates of point A? Show your complete algebraic work.

4 [Foundational] • Target: Curve Sketching & Advanced Analysis

The curve $y = x^2$ is translated by the vector $(-7; -2)$. State the coordinates of the new vertex.

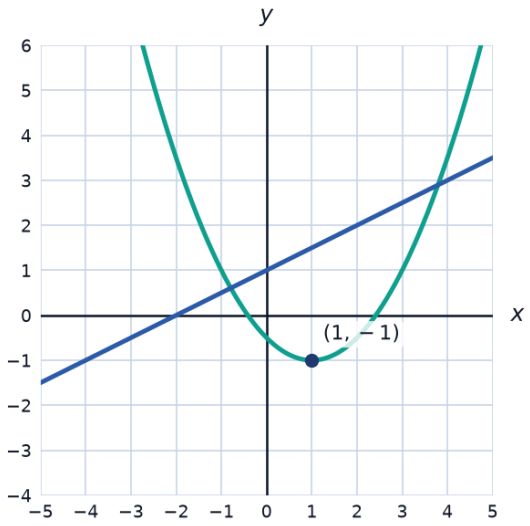
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5 [Foundational] • Target: Basic Graph Forms & Translations

The curve $y = x^2$ is translated by the vector $(-7;\ -1)$. State the coordinates of the new vertex.

6 [Foundational] • Target: Reflections and Stretches

The curve $y = x^2$ is translated by the vector $(-1;\ -2)$. State the coordinates of the new vertex.



7 [Foundational] • Target: Composite Transformations

The minimum point of $y = x^2$ is translated by the vector $(-4;\ -2)$ to give curve C. State the coordinates of the minimum point of C.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Curve Sketching & Advanced Analysis

The minimum point of $y = x^2$ is translated by the vector $(-6;\ -6)$ to give curve C. State the coordinates of the minimum point of C.

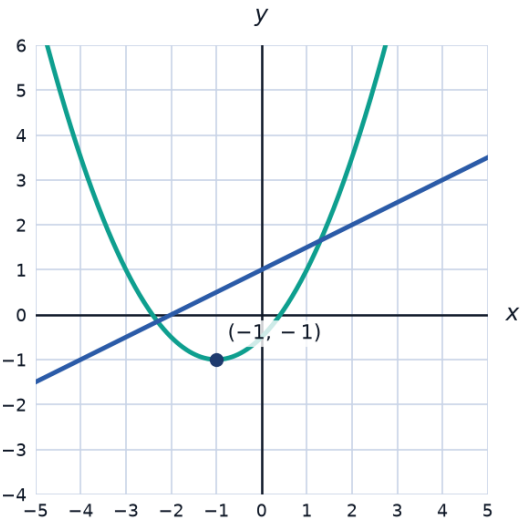
9 [Intermediate] • Target: Basic Graph Forms & Translations

The minimum point of $y = x^2$ is translated by the vector $(-7;\ -3)$ to give curve C. State the coordinates of the minimum point of C.

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10 [Intermediate] • Target: Reflections and Stretches

The minimum point of $y = x^2$ is translated by the vector $(-6;\ -2)$ to give curve C. State the coordinates of the minimum point of C.



11 [Intermediate] • Target: Composite Transformations

The minimum point of $y = x^2$ is translated by the vector $(-4;\ -5)$ to give curve C. State the coordinates of the minimum point of C.

12 [Intermediate] • Target: Curve Sketching & Advanced Analysis

The minimum point of $y = x^2$ is translated by the vector $(-6;\ -3)$ to give curve C. State the coordinates of the minimum point of C.

13 [Intermediate] • Target: Basic Graph Forms & Translations

The minimum point of $y = x^2$ is translated by the vector $(-3;\ -3)$ to give curve C. State the coordinates of the minimum point of C.

14 [Intermediate] • Target: Reflections and Stretches

The curve $y = x^2$ is stretched parallel to the y-axis with scale factor 3, then translated by $(-2;\ -6)$. State the coordinates of the new vertex.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

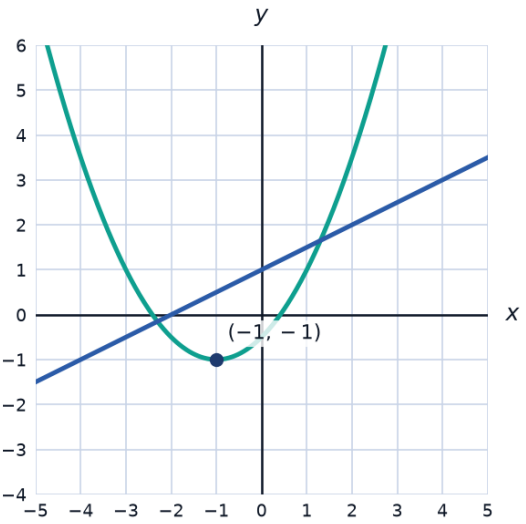
15 [Challenging] • Target: Composite Transformations • Cross-Lesson: 17.01 Discrete Random Variables

The curve $y = x^2$ is stretched parallel to the y-axis with scale factor 2, then translated by $(-6;\ -5)$. State the coordinates of the new vertex.

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16 [Challenging] • Target: Curve Sketching & Advanced Analysis • Cross-Lesson: 18.01 Binomial Distribution

The curve $y = x^2$ is stretched parallel to the y-axis with scale factor 3, then translated by $(-3;\ -1)$. State the coordinates of the new vertex.



17 [Challenging] • Target: Basic Graph Forms & Translations • Cross-Lesson: 19.01 Normal Distribution

The curve $y = x^2$ is stretched parallel to the y-axis with scale factor 3, then translated by $(-5;\ -6)$. State the coordinates of the new vertex.

18 [Challenging] • Target: Reflections and Stretches • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

The curve $y = x^2$ is stretched parallel to the y-axis with scale factor 3, then translated by $(-3;\ -5)$. State the coordinates of the new vertex.

19 [Challenging] • Target: Composite Transformations • Cross-Lesson: 4.01 Coordinate Geometry

The curve $y = x^2$ is stretched parallel to the y-axis with scale factor 3, then translated by $(-2;\ -2)$. State the coordinates of the new vertex.

20 [Challenging] • Target: Curve Sketching & Advanced Analysis • Cross-Lesson: 5.01 Trigonometry

The curve $y = x^2$ is stretched parallel to the y-axis with scale factor 3, then translated by $(-6;\ -2)$. State the coordinates of the new vertex.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Basic Graph Forms & Translations

Describe the transformation that maps the curve $y = f(x)$ onto $y = f(x - 3) + 5$.

a. Translation by vector $[3, 5]$

b. Translation by vector $[-3, 5]$

c. Translation by vector $[3, -5]$

d. Translation by vector $[-3, -5]$

22 [Foundational] • Target: Basic Graph Forms & Translations

The curve $y = x^2$ is translated by vector $[-2, -4]$. What is the equation of the resulting curve?

a. $y = (x + 2)^2 - 4$

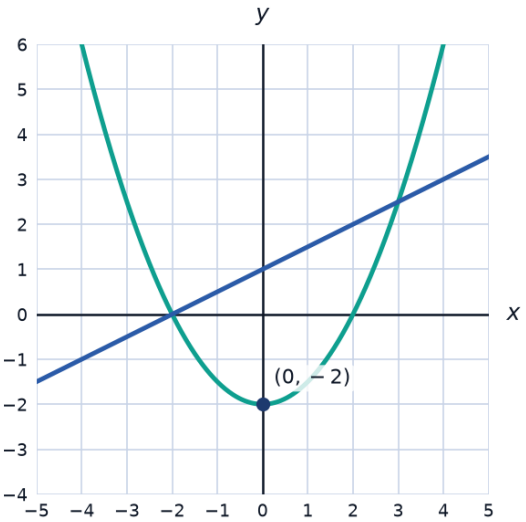
b. $y = (x - 2)^2 - 4$

c. $y = (x + 2)^2 + 4$

d. $y = x^2 - 2x - 4$

23 [Foundational] • Target: Basic Graph Forms & Translations

Find the coordinates of the turning point of $y = 4 - (x - 1)^2$.



a. $(1, 4)$

b. $(-1, 4)$

c. $(1, -4)$

d. $(4, 1)$

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24 [Foundational] • Target: Reflections and Stretches

Describe the transformation that maps $y = f(x)$ onto $y = -f(2x)$.

a. Reflection in x-axis followed by horizontal stretch scale factor $1/2$

b. Reflection in y-axis followed by vertical stretch scale factor 2

c. Reflection in x-axis followed by horizontal stretch scale factor 2

d. Reflection in y-axis followed by horizontal stretch scale factor $1/2$

25 [Foundational] • Target: Reflections and Stretches

The point $(4, 12)$ lies on the curve $y = f(x)$. What are the coordinates of the corresponding point on $y = 3f(x/2)$?

a. $(8, 36)$

b. $(2, 36)$

c. $(8, 4)$

d. $(2, 4)$

26 [Foundational] • Target: Reflections and Stretches

Which equation represents the reflection of the curve $y = 2^x$ in the y-axis?

a. $y = 2^{-x}$

b. $y = -2^x$

c. $y = -2^{-x}$

d. $y = \log_2(x)$

27 [Foundational] • Target: Composite Transformations

The graph of $y = x^3$ undergoes a stretch scale factor 2 parallel to the y-axis, followed by a translation by vector $[1, -3]$. Find its equation.

a. $y = 2(x - 1)^3 - 3$

b. $y = (2x - 1)^3 - 3$

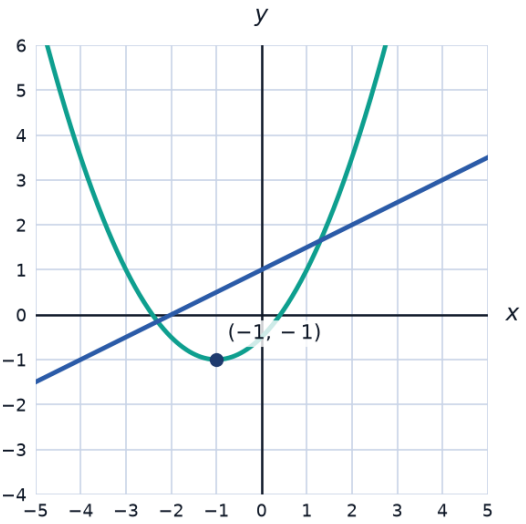
c. $y = 2(x + 1)^3 - 3$

d. $y = 2x^3 - 5$

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28 [Foundational] • Target: Composite Transformations

A curve with equation $y = g(x)$ has a local minimum at $(3, -5)$. Where is the corresponding local extremum on the curve $y = -2g(x + 1)$?



a. Local maximum at $(2, 10)$

b. Local minimum at $(4, 10)$

c. Local maximum at $(4, -10)$

d. Local minimum at $(2, -10)$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Composite Transformations

Find the asymptotes of the rational function $y = \frac{2x+1}{x-3}$.

a. $x = 3$ and $y = 2$

b. $x = -3$ and $y = 2$

c. $x = 3$ and $y = 1$

d. $x = 1/2$ and $y = 3$

30 [Intermediate] • Target: Curve Sketching & Advanced Analysis

Find the coordinates of all intersection points of $y = (x - 2)^2(x + 1)$ with the coordinate axes.

a. x-intercepts at $(2, 0)$ and $(-1, 0)$; y-intercept at $(0, 4)$

b. x-intercepts at $(-2, 0)$ and $(1, 0)$; y-intercept at $(0, -4)$

c. x-intercept at $(2, 0)$; y-intercept at $(0, 4)$

d. x-intercepts at $(2, 0)$ and $(-1, 0)$; y-intercept at $(0, -2)$

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31 [Intermediate] • Target: Curve Sketching & Advanced Analysis

The curve $y = \frac{a}{x+b}$ passes through $(1, 6)$ and $(4, 2)$. Find the values of a and b.

a. $a = 12, b = 1$

b. $a = 6, b = 0$

c. $a = 18, b = 2$

d. $a = 8, b = 0$

32 [Intermediate] • Target: Curve Sketching & Advanced Analysis

Determine the number of real solutions to the equation $|x^2 - 4| = 3$.

a. 4 solutions

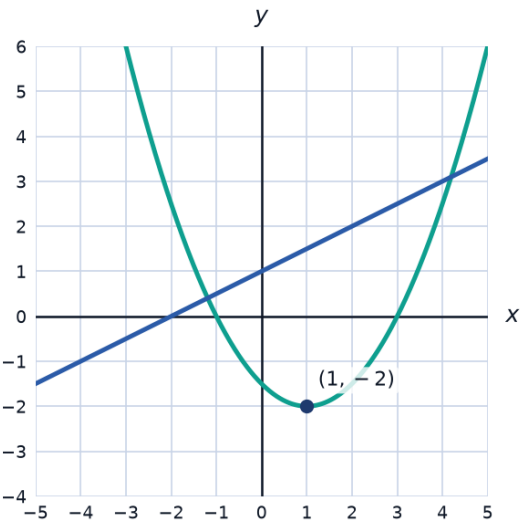
b. 2 solutions

c. 3 solutions

d. 0 solutions

33 [Intermediate] • Target: Basic Graph Forms & Translations

The curve $y = x^2$ is translated by the vector $(4;\ -6)$. State the coordinates of the new vertex.



a. (4,-6)

b. (-4,-6)

c. (4,6)

d. (-6,4)

34 [Intermediate] • Target: Reflections and Stretches

The curve $y = x^2$ is translated by the vector $(-6; 0)$. State the coordinates of the new vertex.

a. (6,0)

b. (0,-6)

c. (-5,0)

d. (-6,0)

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35 [Intermediate] • Target: Composite Transformations

The curve $y = x^2$ is translated by the vector $(6;\ -7)$. State the coordinates of the new vertex.

a. (-6,-7)

b. (6,7)

c. (-7,6)

d. (6,-7)

36 [Intermediate] • Target: Curve Sketching & Advanced Analysis

The curve $y = x^2$ is translated by the vector $(-5; 2)$. State the coordinates of the new vertex.

a. (5,2)

b. (-5,-2)

c. (-5,2)

d. (2,-5)

37 [Intermediate] • Target: Basic Graph Forms & Translations

The curve $y = x^2$ is translated by the vector $(-3; 6)$. State the coordinates of the new vertex.

a. (3,6)

b. (6,-3)

c. (-3,-6)

d. (-3,6)

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Reflections and Stretches • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

The curve $y = x^2$ is translated by the vector $(-4; 2)$. State the coordinates of the new vertex.

a. (-4,-2)

b. (2,-4)

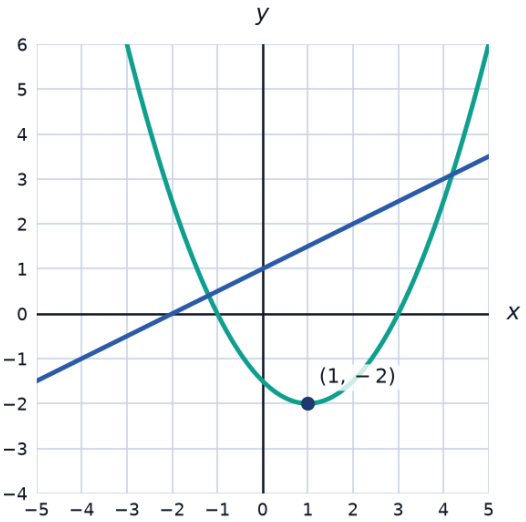
c. (-4,2)

d. (4,2)

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39 [Challenging] • Target: Composite Transformations • Cross-Lesson: 4.01 Coordinate Geometry

The curve $y = x^2$ is translated by the vector $(4;\ -5)$. State the coordinates of the new vertex.



a. (4,-5)

b. (-5,4)

c. (-4,-5)

d. (4,5)

40 [Challenging] • Target: Curve Sketching & Advanced Analysis • Cross-Lesson: 5.01 Trigonometry

The curve $y = x^2$ is translated by the vector $(-7; 3)$. State the coordinates of the new vertex.

a. (-7,-3)

b. (3,-7)

c. (7,3)

d. (-7,3)

41 [Challenging] • Target: Basic Graph Forms & Translations • Cross-Lesson: 6.01 Differentiation & Integration

The curve $y = x^2$ is translated by the vector $(2;\ -4)$. State the coordinates of the new vertex.

a. (-4,2)

b. (-2,-4)

c. (2,-4)

d. (2,4)

42 [Challenging] • Target: Reflections and Stretches • Cross-Lesson: 7.01 Statistics & Probability

The curve $y = x^2$ is translated by the vector $(3;\ -3)$. State the coordinates of the new vertex.

a. (-3,-3)

b. (3,3)

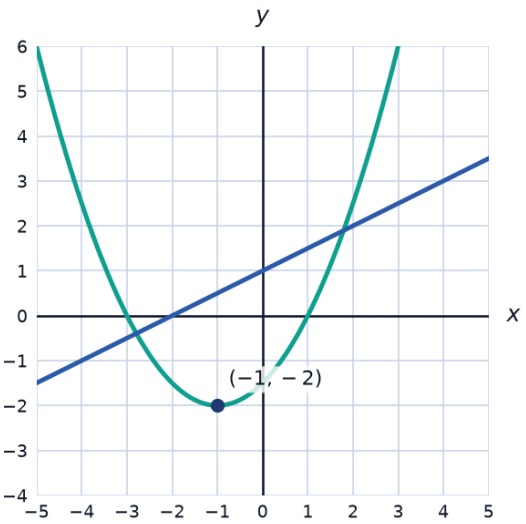
c. (-3,3)

d. (3,-3)

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43 [Challenging] • Target: Composite Transformations • Cross-Lesson: 8.01 Functions: Composite & Inverse

The curve $y = x^2$ is translated by the vector $(5;\ -3)$. State the coordinates of the new vertex.



a. (5,-3)

b. (5,3)

c. (-5,-3)

d. (-3,5)

44 [Challenging] • Target: Curve Sketching & Advanced Analysis • Cross-Lesson: 9.01 Circular Measure

The curve $y = x^2$ is translated by the vector $(-1; 5)$. State the coordinates of the new vertex.

a. (5,-1)

b. (1,5)

c. (-1,5)

d. (-1,-5)

45 [Challenging] • Target: Basic Graph Forms & Translations • Cross-Lesson: 1.01 Indices, Surds & Quadratics

The curve $y = x^2$ is translated by the vector $(1;\ -3)$. State the coordinates of the new vertex.

a. (1,-3)

b. (-1,-3)

c. (-3,1)

d. (1,3)

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